

Question

NH_4HS is a very unstable compound and easily decomposes into NH_3 and H_2S . Put $30.00\text{ g } NH_4HS$ and $0.100\text{ mol } N_2$ into a container with variable volume, then heat to 308.0 K and adjust the external pressure to $P = 2.000 \times 10^5\text{ Pa}$. The decomposition of NH_3 and H_2S can be ignored at this temperature. In this problem, assume that the gases separately obey their own van der Waals equation of state, ignore the volume change caused by mixing, and $p^\circ = 101325\text{ Pa}$. Analyze the deviation calculated using the van der Waals equation and the ideal gas equation. Select the correct one.

	$\Delta_f H_{298}^\circ, \text{ kJ/mol}$	$S_{298}^\circ, \text{ J/(mol} \cdot \text{K)}$
$NH_4HS(s)$	-156.9	113.4
$NH_3(g)$	-46.11	192.45
$H_2S(g)$	-20.63	205.79

	$10^3 a / (\text{Pa} \cdot \text{m}^6 \cdot \text{mol}^{-2})$	$10^6 b / (\text{m}^3 \cdot \text{mol}^{-1})$
$NH_3(g)$	422.5	37.07
$H_2S(g)$	449.0	42.87
$N_2(g)$	140.8	39.13

A. If the ideal gas equation is used, the absolute error in calculating the remaining solid mass is on the order of 10^{-1} g .

B. If using the ideal gas equation, the absolute error in calculating the remaining solid mass is on the order of 10^{-2} g .

- C. If using the ideal gas law, the absolute error in calculating the remaining solid mass is on the order of $10^{-3}g$.
- D. Remaining solid mass $25.55g$
- E. Remaining solid mass $30.00g$
- F. Equilibrium volume $3.51L$
- G. Equilibrium volume $1.28L$
- H. Total amount of gaseous substances produced by the reaction $0.0855mol$
- I. All other options are incorrect.

Answer

Correct Answer: **B**

Detailed Explanation

At 298K, the enthalpy change of the reaction $\Delta_r H_{298}^\circ = (-46.11) + (-20.63) - (-156.9) = 90.16 \text{ kJ/mol}$, entropy change $\Delta_r S_{298}^\circ = 192.45 + 205.79 - 113.4 = 284.84 \text{ J/(mol} \cdot \text{K)}$

CHECKPOINT

0.5 PTS

$$\Delta_r H_{298}^\circ = 90.16 \text{ kJ/mol}, \Delta_r S_{298}^\circ = 284.8 \text{ J/(mol} \cdot \text{K)}$$

Then the Gibbs free energy change can be calculated as $\Delta_r G_{298}^\circ = \Delta_r H_{298}^\circ - T\Delta_r S_{298}^\circ = 90.16 - 298.0 \times 284.84 \times 10^{-3} = 5.27768 \text{ kJ/mol}$

CHECKPOINT

0.5 PTS

$$\Delta_r G_{298}^\circ = 5.278 \text{ kJ/mol}$$

From the formula $\Delta_r G_{298}^\circ = -RT \ln K_a^{298K}$ solve for $K_a^{298K} = 0.11881563281$

Assume that the change in $\Delta_r H^\circ$ at 298K and 308K can be ignored, that is $\Delta_r H_{308}^\circ = \Delta_r H_{298}^\circ = 90.16 \text{ kJ/mol}$

CHECKPOINT

0.5 PTS

$$\Delta_r H_{308}^\circ = 90.16 \text{ kJ/mol}$$

To calculate K_p , first calculate K_a . From the formula $d \ln K_a = \frac{\Delta_r H^\circ}{RT^2}$ we get $\ln \frac{K_a^{T_2}}{K_a^{T_1}} = -\frac{\Delta_r H^\circ}{R} \left(\frac{1}{T_2} - \frac{1}{T_1} \right)$, solve for $K_a^{308K} = 0.38725431767$

CHECKPOINT

0.5 PTS

$$K_a^{308K} = 0.3873$$

Let the total volume be V , and the amount of substance converted by the reaction be n_{gas} . Substituting the van der Waals equation for the three gases respectively and rearranging, we have the system of equations

$$\begin{cases} (P(N_2) + \frac{1.408 \times 10^{-3}}{V^2})(V - 3.913 \times 10^{-6}) = 2.560712 \times 10^2 \\ (P(NH_3) + \frac{0.4225 \times n_{gas}^2}{V^2})(V - 3.707 \times 10^{-5} \times n_{gas}) = n_{gas} \times 2.560712 \times 10^3 \\ (P(H_2S) + \frac{0.4490 \times n_{gas}^2}{V^2})(V - 4.287 \times 10^{-5} \times n_{gas}) = n_{gas} \times 2.560712 \times 10^3 \end{cases}$$

And the equilibrium constant satisfies $K_a^{308K} = P(NH_3) \times P(H_2S)/p^{\circ 2}$, and the total pressure satisfies $P = P(N_2) + P(NH_3) + P(H_2S)$.

$$\text{Solve for } \begin{cases} P(N_2) = 7.39225036 \times 10^4 Pa \\ P(NH_3) = 6.30422915 \times 10^4 Pa \\ P(H_2S) = 6.30352049 \times 10^4 Pa \\ n_{gas} = 0.0855127459 mol \\ V = 3.4624675 L \end{cases}$$

CHECKPOINT

1 PTS

Equilibrium volume $V = 3.46L$, options F and G are incorrect

The total amount of gaseous substance produced is $n = 2n_{gas} = 0.171\text{mol}$.

CHECKPOINT

1 PTS

The total amount of gaseous substance produced is $n = 2n_{gas} = 0.171\text{mol}$, option H is incorrect

The mass of the remaining solid is $m = 30.00 - 0.0855127459 \times 51.11244 = 25.62923490595\text{ g}$

CHECKPOINT

1 PTS

The mass of the remaining solid is $m = 25.63\text{ g}$, options D and E are incorrect

Using the ideal gas law to derive the system of equations

$$\begin{cases} P(N_2) = \frac{0.1 \times 8.314 \times 308}{V} \\ P(NH_3) = \frac{n_{gas} \times 8.314 \times 308}{V} \\ P(H_2S) = \frac{n_{gas} \times 8.314 \times 308}{V} \\ 2.000 \times 10^5 = P(N_2) + P(NH_3) + P(H_2S) \\ 0.38725431767 = P(NH_3) \times P(H_2S) / 101325^2 \end{cases}$$

Solve for

$$\begin{cases} P(N_2) = 7.39225038 \times 10^4 Pa \\ P(NH_3) = P(H_2S) = 6.30387481 \times 10^4 Pa \\ n_{gas} = 0.0852768018 mol \\ V = 3.46404930 L \end{cases}$$

The mass of the remaining solid is $m = 30.00 - 0.0852768018 \times 51.11244 = 25.64129799568\text{ g}$

Deviation $m_{ideal} - m_{Van} = 25.64129799568\text{ g} - 25.62923490595\text{ g} = 0.0120630897\text{ g}$

CHECKPOINT

1 PTS

The absolute error is about $0.01g$, which is on the order of $10^{-2}g$, option B is correct, options A and C are incorrect